# Oyster and shiitake mushrooms (cultivated Pleurotus and Lentinula, cadmium accumulators),: HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified standard for Oyster and shiitake mushrooms (cultivated Pleurotus and Lentinula: cadmium accumulators) (Edible Fungi), stated as sold. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; regulatory-alignment / read-across / not-material / control)

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 300 | government maximum level |
| Cadmium (Cd) | Tier 1 | 150 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 300 | government maximum level |

| | | | |
|---|---|---|---|
| Inorganic arsenic (iAs) | Tier 1 | 60 | government maximum level |
| Mercury, total (tHg) | Tier 1 | 30 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 400 | human-grade read-across (European Union) |
| Tin (Sn) | Tier 2 | 250,000 | human-grade read-across (Singapore) |
| Aluminium (Al) | Tier 2 | 100,000 | human-grade read-across (China) |
| Total chromium (Cr) | Tier 2 | 1,000 | human-grade read-across (China) |
| Hexavalent chromium (Cr-VI) | Tier 2 | ≤ Cr | bounded by Cr (reflex speciation) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 300 | government maximum level |
| Cadmium (Cd) | 150 | government maximum level |
| Total arsenic (tAs) | 300 | government maximum level |
| Inorganic arsenic (iAs) | 60 | government maximum level |
| Mercury, total (tHg) | 30 | government maximum level |
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 400 | European Union — human-food analog |
| Tin (Sn) | 250,000 | Singapore — human-food analog |
| Aluminium (Al) | 100,000 | China — human-food analog |

| | | |
|---|---|---|
| Total chromium (Cr) | 1,000 | China — human-food analog |
| Hexavalent chromium (Cr-VI) | ≤ Cr | Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell. |

## 4 | Cite this standard

Pendergrass, K. "Oyster and shiitake mushrooms (cultivated Pleurotus and Lentinula, cadmium accumulators),." In Heavy Metal Tested & Certified (HMTc) Edible Fungi standard, version 1.0 (2026-08-14). Institute of Contaminant Standards (ICS), 2026.

## References


[1] China — human-food maximum levels
Binds Al, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL


### STATUS AND LICENCE


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### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/cultivated--mushrooms-oyster-shiitake. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

### PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-cultivated-mushrooms-oyster-shiitake.